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"GPSBASEDTOLLCOLLECTIONSYSTEM"

issubmittedtoMalnadCollegeofEngineering,Hassan,duringtheacademic year 2024-25 in partial fulfillment for the award of the degree of

BachelorofEngineering In Electronics&CommunicationEngineeringby

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2024-25

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in partial fulfillment for the award of

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Of

Malnad College of Engineering Affiliated

to

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during the year 2023-24.It is certified that all corrections/suggestions indicated for internal Assessment have been incorporate in the project report deposited in the Department Library. The Project report has been approved,as it satisfies the academic requirements in respectof Project Work prescribed for the Bachelor of Engineering Degree.

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DECLARATION

We,**AmruthaBT(4MC21EC006),Bhavya(4MC21EC015),NithinBR(4MC21EC069),Vaibhav R V (4MC21EC107)** students of eighth semester, B.E Electronics and Communication Engineering of Malnad College of Engineering, Hassan, hereby declare that the main project work titled **"GPS BASED TOLLCOLLECTIONSYSTEM"**hasbeencarriedoutbyusundertheguidanceof**Dr.PrakruthiH.L**,Associate Professor, Department of Electronics & Communication Engineering, Malnad College of Engineering, Hassan in the partialfulfillmentofrequirementsfortheawardofdegreeinBachelorofEngineeringElectronicsandCommunication Engineering of Malnad College of Engineering, Hassan under Visvesvaraya Technological University, Belgaum during the academic year 2024-2025.

Wealsodeclarethat,tothebestofourknowledgeandbelief,thematterembodiedinthisprojecthas not been submitted previously by us for the award of any degree or to any other University.

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ABSTRACT

The increasing demand for efficient transportation infrastructure has highlighted the limitations of conventional toll collection systems, which often rely on physical checkpoints and manual processing. This project presents a smart, GPS-based automated toll collection system that leverages the ESP32 microcontroller, NEO-6M GPS module, and wireless communication technologies to calculate toll charges dynamicallybasedonreal-timevehiclemovement.Thesystemcontinuouslymonitorslocationdatausingthe GPS module and calculates the distance traveled through the Haversine formula. To ensure accuracy, a movementthresholdisimplementedtofilteroutnegligibleGPSdrift.Onceapredefineddistancethresholdis reached,theESP32communicateswiththeTwilioAPIoverWi-Fito sendanSMSalerttotheuser,detailing the total distance covered and the corresponding toll amount.

Theproposed systemeliminatestheneed fortraditionaltollbooths,therebyreducingcongestion,minimizing humanerror,and improvingcost-efficiency.Real-worldtesting confirmed the system's reliabilityintracking motion,calculatingtollsaccurately,anddeliveringtimelynotifications.Theprojectdemonstrateshowmodern microcontroller platforms can be used to create scalable and flexible tolling systems that support real-time communication and billing. It lays the groundwork for future enhancements such as GSM-based messaging, mobile app integration, and cloud data storage. This automated toll solution offers a practical and innovative approach toward smart transportation and the development of intelligent road infrastructure.

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CHAPTER1

INTRODUCTION

With therapid expansion ofintelligent transportation systems, theneed forautomatedand efficient toll collection mechanisms has grown significantly. Traditional toll booths often lead to congestion, increased travel time, and fuel consumption. To address these challenges, this project introduces aGPS-based automated toll collection system built aroundthe ESP32 microcontroller. The system is designed to monitor vehicle movement in real-time using geolocation data and automatically compute toll charges based on the distance traveled. Notifications regarding toll updates are sent directly to the user's mobile phone through SMS, ensuring seamless communication and transparency.

At the core of this system lies the ESP32, a powerful and versatile microcontroller with built-inWi-Ficapabilities,whichactsasthe centralunitforprocessingand control.TheNEO-6M GPS module is employed to gather accurate location data, while the SIM800L GSM module facilitates communication via SMS. The distance between the starting point and the current positioniscomputedusingtheHaversineformula,whichprovidesprecisemeasurementsbasedon latitude and longitude. To avoid unnecessary computations and reduce noise due to GPS fluctuations,thesystemincludesaminimummovementthreshold.Onlysignificantmovements—defined as changes exceeding 0.5 meters—are considered for distance tracking.

Thetollamountisdynamicallycalculatedby multiplyingthe totaldistance(inkilometers) withapredefinedrate.Oncethevehiclecoversaspecifieddistancethreshold(e.g.,10kilometers), an SMS containing the distance traveled and the calculated toll is sent to a registered recipient number. This approach not only automates toll computation but also offers real-time updates without the need for manual interaction or infrastructure-heavy solutions like RFID or ANPR systems.

The implementation further ensures reliability through continuous status monitoring of all modules. LEDs on the GPS and GSM modules indicate operational readiness, and the system includes safety checks to confirm all components are functioning before use. This design is particularly useful in areas where conventional toll systems are impractical, enabling a low-cost and scalable alternative suitable for rural or mobile deployment scenarios.

Overall, this project demonstrates a practical and innovative solution for modernizing toll collection systems using embedded technology and wireless communication. It exemplifies how IoT-enabled devices can transform traditional transportation infrastructure into smarter, more efficient networks.

Aboutproject

The project titled **"GPS Based Toll Collection System"** focuses on modernizing traditional toll collection methods by introducing an automated, infrastructure-free solution using GPStechnology.ThissystemutilizestheESP32microcontrollerandtheNEO-6MGPSmoduleto continuouslymonitorthevehicle'sreal-timelocationandcomputetollchargesbasedontheactual distance traveled.

Thecoreofthesystemisbuiltaroundthe**Haversineformula**,whichaccuratelycalculates the distance between two GPS coordinates. By applying a predefined toll rate per kilometer, the systemdynamicallyupdatesthetollamountasthevehiclemoves.Aminimummovementthreshold is incorporated to ignore minor GPS fluctuations and ensure reliable distance tracking.

This approach eliminates the need for manual toll booths or RFID checkpoints, reducing traffic congestion, travel delays, and operational costs. The compact and portable nature of the system makes it ideal for both urban and rural deployment, especially in areas where physical infrastructure is limited.

Overall, the project demonstrates a cost-effective, scalable, and intelligent toll collection method that supports the development of smart transportation systems and contributes to sustainable mobility solutions.

ProblemStatement

Traditional toll collection systems, including manual booths, RFID scanners, and camera-based license plate recognition technologies, are often plagued by inefficiencies such as traffic congestion,highinfrastructurecosts,limitedscalability,andlackofreal-timeusercommunication.
These systems are typically fixed-location, dependent on heavy roadside hardware, and are not practical in rural or remote areas with limited resources. Additionally, many existing systems fail to provide fair, distance-based tolling and offer minimal transparency to end users during travel.

There is a growing need for an intelligent, automated, and infrastructure-independent tolling mechanism that can calculate tolls dynamically based on the actual distance traveled and communicate toll charges directly to users in real time. This solution should be cost-effective, portable,easytodeploy,andcapableofoperatingwithoutrelyingontraditionalphysicaltollbooths or high-maintenance equipment.

This project addresses the problem by developing a GPS-based automated toll collection system using the ESP32 microcontroller, NEO-6M GPS module, and SIM800L GSM communicationforSMSalerts.Thesystemcomputestollchargesbasedongeolocationdata,filters outGPSdriftthroughamovementthreshold,and sendstollupdatesviaSMS,therebyproviding a smart, real-time, and user-friendly tolling solution suitable for modern transportation infrastructure.

Objectives

The primary objective of this project is to design and implement an efficient, real-time GPS-based toll collection system using the ESP32 microcontroller. This system aims to automate theprocessoftollcalculationbasedontheactualdistancetraveledbyavehicle,therebyeliminating the need for conventional toll booths and reducing manual intervention. By leveraging the capabilitiesoftheNEO-6MGPSmodule,thesystemcontinuouslytracksthegeographicalposition of the user in real-time, calculates the distance covered using precise geospatial data, and dynamically determines the toll charges using a fixed rate per kilometer. Another key objective is to enhance user communication by integrating the SIM800L GSM module, which enables the systemtosendtollupdatesviaSMSthroughaweb-basedAPIservicelikeTwilio.Theprojectalso aimstoensureportability,lowcost,andminimalinfrastructuredependency,makingitsuitablefor both urban and rural environments where fixed toll systems may not be feasible. In addition, the system includes safeguards such as movement thresholds to avoid errors due to GPS jitter and incorporates visual diagnostics to confirm module readiness, ensuring reliability during field deployment.Overall,theprojectstrivestodevelopascalable,user-friendly,andautomatedtolling solutionthatreflectstheprinciplesofsmarttransportationandInternetofThings(IoT)integration.

CHAPTER2
LITERATURESURVEY

A toll collection system using GPS was presented in [1] with a vision to enhance toll efficiencyonfreewayswithouttherestrictionsofRFID-basedandmanualtollbooths.Thissystem involves the integration of an NEO 6M GPS module and an Arduino Uno microcontroller for tracking the movement of vehicles. Upon approaching a toll zone, the location of the vehicle is storedbasedonpreviouslysavedGPScoordinates.Thetraveleddistanceisthencalculatedinreal- timeby thesystem based on theHaversineformulaand a fixed rate ofchargepermeter is applied to calculate the toll fee. The results are displayed on a 16x2 LCD display for clarity. The method provides equitable tolling through levying charges only for the actual distance covered and does away with the need for vehicles to be halted at booths.

The[1]designalso considers energyand system efficiency with checkson validGPS data included and fallback error messages when there is not enough signal. The system is coded with the Tiny GPS++ library and aims at low-cost, real-time embedded applications and is well-suited todevelopingnationssuchasIndia.Thewholeprocessoftollingbecomesautomaticandcashless, leading to considerable reductions in fuel consumption, traffic jam, and time wasted in lengthy queues.

In [2], a new GPS-based toll collection model was established based on a Raspberry Pi 2 microcontrollerandothermodulessuchasLCD,Wi-Fi,andspeakers.Incontrastto[1],thismodel incorporates online database connectivity through a personal cloud server that provides remote accessandmonitoringoftollrecords.Travelhistory,speedingviolations,andtolltransactionsare allstoredinstructuredSQLdatabases.ThePython-basedsystemarchitectureincludesGPSsignal detection,real-timeupdatesontollcharges,andspeedmonitoring—makingitsuitablenotonlyfor toll collection but also for traffic law enforcement.

A notable innovation in [2] is the system's ability to detect loss of GPS or internet connectivity. By pinging online servers and checking GPS signal variables, the system ensures robustnessinoperation.Inaddition,thereisanautomaticdelaytimeadjustmentfunctionthatsaves powerthroughmodulatingdataacquisitionfrequencybasedonvehiclespeed.Thedeploymentwas trialedonMalaysianhighways,wherethetestexhibitedhighaccuracyandefficiency.Theaddition of cloud-based access provides remote oversight abilities, a key feature in future smart infrastructure.

The study in [3] investigates an IoT smart gate system with Node MCU ESP8266, ultrasonicsensors,andcameramodulesformonitoringvehicleentryinresidentialandeducational environments. The system identifies vehicle presence automatically, captures images of license plates,andcross-verifiesthemagainstaclouddatabasebeforeopeningthegates.SMSnotifications are sent for invalid attempts, and administrators can remotely open gates via a web interface. The emphasis here is less on toll computation and security, but more on access control, and yet the principles intermingle with tolling structures, particularly in multi-zone lots or gated campuses.

[3] also describes the integration of the system with machine learning algorithms for anomaly detection, thus making it more adaptive to changing access patterns. Scalability is made possible by cloud services, and technologies such as biometric authentication and RFID enable multi-factor authentication. Although robust, the paper identifies challenges such as high installationcosts,difficultyinintegratingwithlegacysystems,anddataprivacyissues—factorsof note for tolling systems that intend to scale nationwide.

In [4], the Smart Gate System continues to leverage real-time analytics and adds sophisticated data visualization using dashboards. The architecture of the system consists of three core layers: perception (sensing of data through sensors), network (transmission of data), and application (analysis of data and user interface). This layered structure provides effective processing and facilitates predictive capabilities like traffic prediction. Edge computing further reduces latency—of paramount importance in high-volume toll zones.

Oneofthestrengthsemphasizedin[4]isincreasedsecuritywithencryptedcommunication and block chain integration to ensure tamper-proof records. These capabilities facilitate deployment in high-risk environments and render the system fraud-proof or resistant to unauthorized access. The system was tested in a college campus environment with encouraging results for usability, speed, and accuracy.

Collectively, these systems represent a trend toward smart toll and access management, focusingonautomation,visibility,andleanphysicalinfrastructure.GPSandIoTtechnologiesoffer a cost-efficient and scalable substitute for legacy gantry and RFID solutions. Vehicle position tracking, cloud storage, and real-time user engagement create the foundation of current smart mobility applications.

Buteverysystemhas its compromises.GPS-basedsolutionssuch asin [1] and[2]provide distance-basedaccuracyoftollingbutrelyonsatellitevisibilityandconnectivity.Atthesametime, IoT-basedgatesystems in[3]and[4]providestrongersecurityand remote-access capabilities but are potentially more complex and costly to implement. The perfect future system would integrate GPS tracking, RFID fallback, and real-time cloud analytics into a strong, multi-layered tolling architecture.

Insummary,thepapersasawholehighlighttheneedtointegrateembeddedsystems,GPS, IoT,andcloudcomputingtosolvetheinefficienciesofconventionaltolling.Frommicrocontroller- based embedded solutions to cloud-enabled smart gates, each solution has its own strengths. The direction of this technology is toward smooth, automated tolling and access control solutions that can be integrated into smart city infrastructure with minimal human intervention.

CHAPTER3

RESEARCHGAPSOFEXISTINGMETHODS

Despite the advancements in digital toll collection systems, many existing solutions still facelimitations intermsofaccessibility, scalability, andefficiency. Thetraditional methods,such as manual toll booths, RFID-based systems, and camera-based license plate recognition (ANPR), while effective to a degree, present several shortcomings that highlight the need for more innovative and flexible solutions. The following research gaps have been identified in the context of existing toll collection methods:

1. InfrastructureDependency

Most current toll systems rely heavily on fixed roadside infrastructure, such as toll plazas, RFIDreaders,orhigh-resolutioncameras.Theseinstallationsareexpensivetosetupandmaintain, and they are generally limited to major highwaysand urban centers. This makes them impractical for rural or less-traveled roads where the cost cannot be justified. There is a clear gap in the availability of lightweight, portable, and infrastructure-independent systems that can operate autonomously.

2. HighImplementationandMaintenanceCosts

RFID-based and ANPR systems require complex hardware and software integration, including sensors, barriers, and centralized servers. These setups involve significant capital investment and periodic maintenance to ensure accuracy and reliability. In contrast, a GPS-based model using mobile communication like GSM presents a cost-effective alternative, but has yet to be widely adopted or optimized.



Figure1.ESP32GPSTracker-IoTbasedVehicleTracking System

3. TrafficCongestion andDelays

Manual toll booths and semi-automated systems often contribute to traffic congestion, especially during peak hours. Vehicles must slow down or stop to pay tolls or be scanned by sensors. This delay not only frustrates commuters but also increases fuel consumption and emissions. The need for a frictionless, on-the-move toll collection mechanism remains largely unmet.

4. LimitedReal-TimeUserInteraction

Existing systems seldom provide real-time toll updates to users. In most cases, users are unaware of the toll amount until after they pass a toll point or receive a summary at the end of a journey.Thereisagapinreal-timecommunicationanduserawarenessthatcouldbebridgedusing technologies such as SMS notifications or mobile app alerts integrated with GPS tracking.

5. RestrictedCoverageand Scalability

RFID tags are only effective within a limited range and are restricted to predefined checkpoints. Similarly, camera-based systems may struggle in low-visibility conditions or with number plate obstructions. These constraints limit the scalability of such systems across varied geographic and climatic conditions. A GPS-based solution, being satellite-driven, offers global coverageand can adapttoawiderrangeofenvironmentsbuthasnotbeen extensivelyexploredin toll collection scenarios.

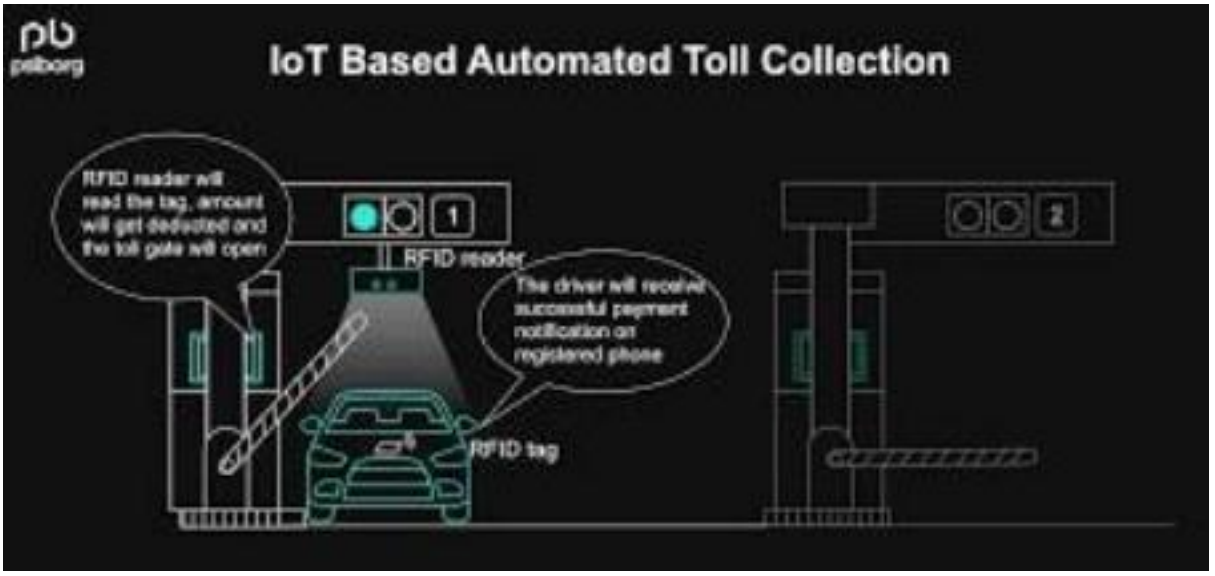


Figure2.IoTBased AutomatedTollCollection

6. PrivacyandDataSecurityConcerns

While modern systems collect vast amounts of location and identity data, they often lack robust privacy frameworks. This raises concerns about how user data is stored, transmitted, and potentially misused. Systems using direct communication methods like SMS need to incorporate secure authentication and encryption measures, a gap often overlooked in prototype-level implementations.

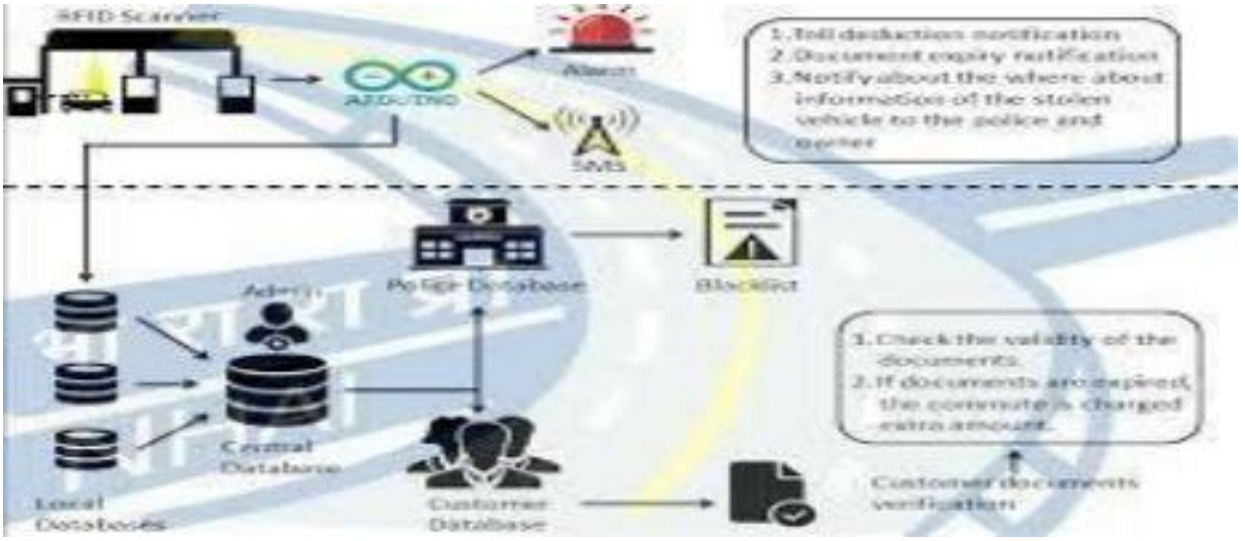


Figure3.BlockDiagramofAutomated TollCollection

7. LackofFlexibilityforDynamicPricingModels

Most existing toll systems operate on static pricing models based on fixed checkpoints. They are not designed to handle dynamic tolling based on actual distance traveled or congestion levels. A GPS-driven system, however, offers the potential for real-time, usage-based tolling models,butfurtherresearch anddevelopment areneededtomakesuch systems reliable,accurate, and widely acceptable

Thisanalysisofresearchgapsrevealsthelimitationsincurrenttollsystemsandunderscores thepotentialofGPS-andGSM-basedmodels,suchastheoneimplementedintheproject,tooffer a more adaptable, cost-effective, and user-centric approach to toll collection.

CHAPTER4
TOOLSANDTECHNOLOGYUSED

The project integrates a range of hardware components, development platforms, programming tools, and APIs to design and implement a real-time, automated toll collection system. The key tools and technologies utilized are as follows:

HardwareComponents

- **ESP32Microcontroller**:Actsasthecentralprocessingunit;handlesGPSdataacquisition, toll calculations, and communication tasks.
- **NEO-6MGPSModule**:Providesreal-timegeolocationdata(latitudeandlongitude)for vehicle tracking.
- **SIM800LGSMModule**:EnablesSMScommunicationthroughmobilenetworks.
- **LEDIndicators**:Providevisualfeedback onmodule statusand diagnostics.
- **PowerSupplyModule/USBPowerBank**:Ensuresportablepowerforthesystemduring testing and deployment.

SoftwareandDevelopmentTools

- **ArduinoIDE**:UsedforprogrammingtheESP32anduploadingcode.
- **TinyGPS++Library**:AGPSparsinglibraryforprocessingNMEAdatafromtheGPS module.
- **WiFi.handHTTPClient.hLibraries**:EnableWi-FiconnectionandHTTP-basedcommunication with external servers.
- **ArduinoJsonLibrary**:Usedforconstructing andparsingJSONpayloadsforAPIrequests.
- **SerialMonitor**:Fordebuggingandmonitoringliveg PSdata,distances,andtoll computations.

CommunicationandCloudServices

- **TwilioAPI**:Cloud-basedSMSgatewayusedtosendreal-timetollupdatestotheuser's mobile phone.
- **Wi-FiNetwork(SSID:wifiplane)**:ProvidesinternetaccessfortheESP32tocommunicate with Twilio servers.

MathematicalandLogicalAlgorithms

- **HaversineFormula**:Calculatestheshortestdistancebetweentwogeographicpointson Earth using latitude and longitude.
- **ThresholdFilteringLogic**:Ignoresmovementbelow0.5meterstominimizeGPSnoise and improve accuracy.

CHAPTER5
PROPOSEDMOTHODOLOGY

The proposed methodology outlines the development of a portable, GPS-enabled automated toll collection system using the ESP32 microcontroller. This system is designed to overcome the limitations of conventional toll systems by eliminating the need for fixed infrastructure, providing real-time distance tracking, and notifying users of toll charges through SMS communication. It offers a lightweight and scalable solution suitable for both urban and remote areas. The methodology integrates hardware setup, software logic, and communication protocols to ensure seamless functionality.

1. SystemArchitectureandComponentIntegration

Thesystemarchitectureconsists ofthreekeyhardwaremodules:

- **ESP32Microcontroller**:Servesasthecentralprocessingunit,managingdataacquisition, processing, and communication.
- **NEO-6MGPS Module**:Continuously capturesgeolocationdata(latitudeandlongitude).
- **SIM800LGSMModule**: SendsSMSupdateswithtolldetailstotheuser'smobilephone.

TheESP32isconnectedtoaGPSmoduleviaUART2(usingGPIOpins16and17forRXandTX, respectively),whiletheGSMmodulehandlesSMSdeliverythroughHTTP-basedcommunication (via the Twilio API) over a Wi-Fi network.

2. Wi-FiConfigurationand Connectivity

The ESP32 connects to a predefined Wi-Fi network (SSID: *wifiplane*) at startup. This connection enables the deviceto accessthe internetand interactwiththe TwilioAPItosendSMS messages. A loop ensures that the ESP32 does not proceed until the Wi-Fi connection is successfully established.

3. Real-TimeGPS DataAcquisition

Oncethesystemisonline,theGPSmodulestartsacquiringlocationdata.TheTinyGPS++ library is used to parse and extract valid latitude and longitude values. The system waits until the GPS has a valid fix and begins monitoring positional changes in real-time.

4. DistanceCalculationUsingtheHaversineFormula

To determine the total distance traveled, the system applies the Haversine formula. This mathematical model calculates the shortest distance between two geographical points over the Earth's surface, based on their latitude and longitude. The formula ensures accurate distance measurements, which are then accumulated over time.

Tofilteroutnoiseandavoidtrackingminor GPSfluctuations,amovementthresholdof**0.5 meters** is implemented. Any movement below this threshold is ignored to improve accuracy and stability.

5. TollCalculationLogic

The total distance (in kilometers) is multiplied by a fixed toll rate (₹50 per kilometer, as defined in the code) to compute the current toll charge. This amount dynamically increases as the device continues to track movement.

6. AutomatedTollNotificationviaSMS

Whentheaccumulateddistancereaches**10kilometers**beyondthelastnotificationpoint,the system prepares a toll update message. This message includes:

- Thetotaldistancetraveled(in kilometers)
- Thecorresponding toll charge

The message is sent via SMS using the Twilio HTTP API. The ESP32 uses an HTTP client to format and transmit the message, including appropriate authentication with the Twilio service. A successful HTTP response indicates that the message was sent successfully.

7. SafetyChecksandDiagnostics

Thesystemincorporatesbasicdiagnosticindicators:

- TheGPS module'sredLEDblinkseverysecondwhenavalid signalisacquired.
- TheGSMmoduleexhibitsasimilarblinkpatterntoconfirmreadiness.
- TheESP32'sonboardLEDconfirmspowerandstartupsuccess.

These indicators help users verify module functionality before deployment in outdoor environments.

8. SimulationandReal-World Testing

Duringtesting,userswalkormovewiththedevicetosimulatevehiclemotion.Thesystem continuouslymonitorslocation,calculatesdistance, andsendsSMS notifications whenthresholds are met. This simulation validates the system's effectiveness in real-world conditions, demonstrating how it can be used for mobile toll tracking.

CHAPTER6

SYSTEMDESIGN&IMPLEMENTATION

The system is designed as a compact and portable GPS-based toll collection unit that integrates location tracking, wireless communication, and automated billing functionalities using the ESP32 microcontroller. At the heart of the design is the ESP32, which coordinates the entire operation by interacting with both the GPS and GSM modules while also handling computation and network communication tasks. The hardware configuration includes a NEO-6M GPS module connected to the ESP32 via UART2 forcontinuous acquisition of real-time latitude and longitude data.Totransmittollinformationtousers,aSIM800LGSMmoduleisinterfacedthroughaWi-Fi network using the Twilio API for SMS communication.

Upon initialization, the ESP32 establishes a connection to a predefined Wi-Fi hotspot (SSID: *wifiplane*) to enable internet access required for HTTP-based communication. Once connected, the system begins monitoring GPS data using the TinyGPS++ library to parse and interpret location values. The implementation uses the Haversine formula to calculate the linear distance between two sets of coordinates, representing successive positions of a moving vehicle. Aminimummovementthresholdof0.5metersisusedtofilteroutinsignificantchangesinposition, ensuring only meaningful displacement is considered in distance calculations.

As the vehicle moves, the total distance is incrementally calculated and stored. The toll amountisdynamicallyderivedbymultiplyingthecumulativedistance(inkilometers)byaconstant tollrate.Whenthedistancecrossesadefinedthreshold—suchasevery10kilometers—thesystem automaticallyformatsamessagecontainingthetotaldistancetraveledandthecurrenttollamount. This message is then sent to the user's mobile phone via the Twilio API, using secure HTTP requests handled by the ESP32's HTTPClient library.

Tomaintainoperationalintegrity,thesystemincludesbasicvisualindicators.TheGPSand GSM modules each feature LED indicators that blink at regular intervals to confirm active communication and signal acquisition. These indicators help users verify that all components are functioningcorrectlybeforeandduringoperation.Theimplementationisfurthervalidatedthrough real-world simulation, where users physically move with the device to emulate vehicle motion. Thispracticalapproachallowsforaccuratetestingofthetracking,calculation,andcommunication processes in varied environments.

Overall, the system design emphasizes simplicity, mobility, and effectiveness by minimizing hardware complexity while maximizing functionality. It provides a viable alternative to traditional toll systems by eliminating the need for stationary infrastructure and offering real-time updates through a user-friendly communication method. The successful integration of GPS tracking with SMS notifications makes this system well-suited for deployment in both urban and remote transportation settings.

CHAPTER7

OUTCOMES

ThedevelopedGPS-basedautomatedtollcollectionsystemsuccessfullymeetsitsintended objectivesbydeliveringafunctional,real-timetollmonitoringandnotificationplatformusinglow-costandportablecomponents.Oneofthekeyoutcomesistheaccuratetrackingofdistancetraveled using the NEO-6M GPS module, which provides reliable and continuous location data. Through the implementation of the Haversine formula and an intelligent threshold mechanism, the system effectivelyfiltersoutminorGPSnoiseandcomputesmeaningfulmovement onlywhenthedevice surpasses a set displacement, ensuring precision in distance measurement. This, in turn, results in theaccuratecalculation oftollchargesbasedonactualkilometerstraveledratherthanpre-defined toll points, addressing the limitations of traditional fixed-location systems.

Anothersignificantoutcomeisthesuccessfulintegrationofwireless communicationusing the ESP32's Wi-Fi capabilities and the Twilio API. The system can generate and send SMS messages automatically once a specified distance threshold is crossed, keeping users informed in real-timeaboutthetollchargesincurredduringtheirjourney.Thisenhancestransparencyanduser awareness, which are often lacking in existing toll collection systems. The automated messaging system performed reliably during field tests, confirming that real-time communication over a mobile network is feasible using a microcontroller-based platform.

In terms ofsystemreliabilityandusability,visual feedbackthroughLEDindicatorson the GPS andGSM modulesproved valuableduringsetup anddeployment, helping usersconfirm that eachmodulewasfunctioningcorrectly.Additionally,thedesignprovedtobeenergy-efficientand suitable for mobile use, powered via USB or battery pack, making it adaptable for deployment in various transport scenarios, including private vehicles, taxis, or commercial fleets.

Overall,thesystem demonstrateditspotential asa scalableand infrastructure-independent tolling solution. It provides a strong foundation for further enhancements, such as mobile app integration, dynamic toll rate adjustment, or the addition of offline SMS capabilities using GSM ratherthanWi-Fi.ThesuccessfulimplementationconfirmsthatGPSandwirelesstechnologiescan be combined effectively to create a smart, user-friendly, and real-time toll collection system.

CHAPTER8
RESULTSANDDISCUSSIONS

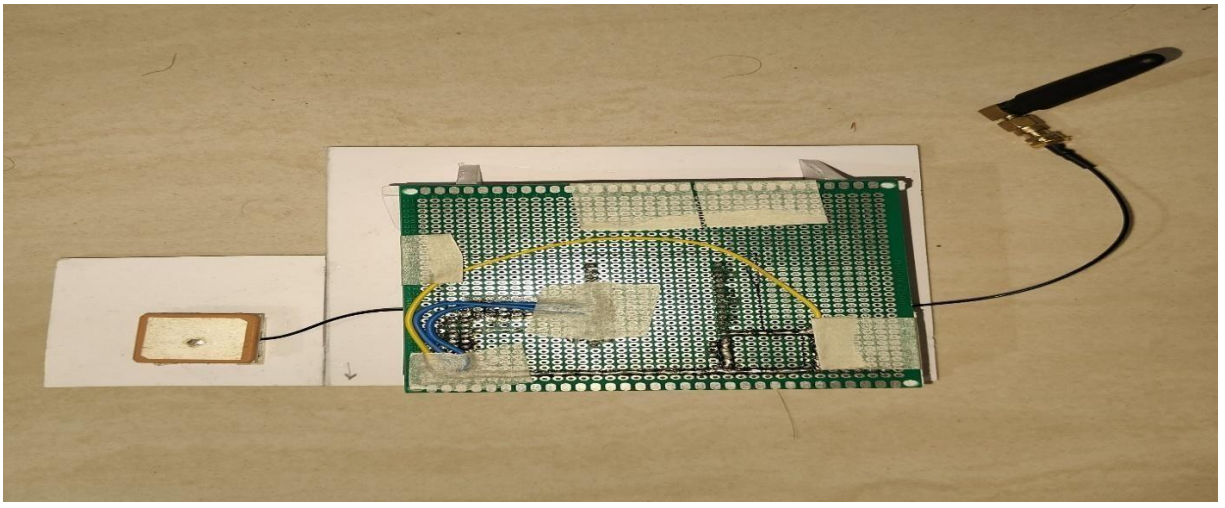
The implementation of the GPS-based automated toll collection system produced effective and consistent results under real-world testing conditions. The system was able to accurately monitor vehicle movement using the NEO-6M GPS module, with GPS data being reliably parsed and updated through the TinyGPS++ library. When the device was moved across different locations—simulating vehicle travel—it successfully calculated the distance between successive GPS coordinates using the Haversine formula. The introduction of a minimum movement threshold of 0.5 meters ensured that only significant positional changes were considered,whichhelped eliminateinaccuraciescausedbyminorGPSsignalfluctuationsordrift.

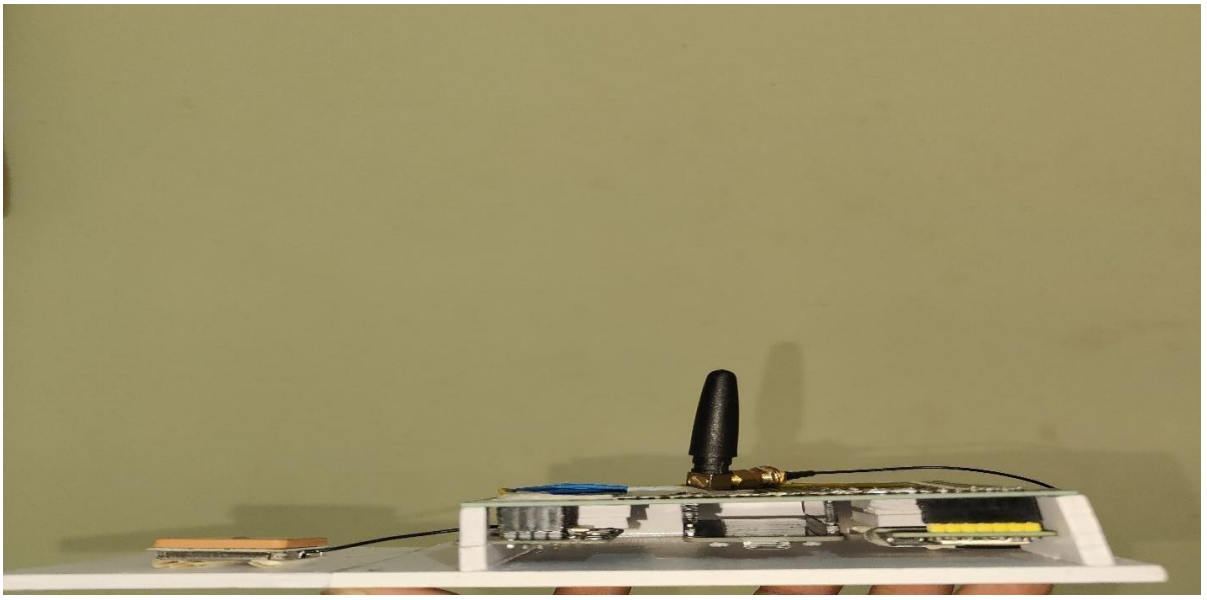
TheESP32microcontrollerprovedtobeacapableplatformforhandlingmultiplefunctions simultaneously, including reading GPS data, managing distance calculations, and communicating with external servers. Once a distance of 10 kilometers was reached from the last SMS trigger point, the system correctly formatted and sent an SMS containing the total distance covered and thecorrespondingtollamount.Thisprocesswassuccessfullyrepeatedateachinterval,confirming the robustness of the distance tracking and notification logic. The integration of the Twilio API allowed for reliable message delivery over Wi-Fi, with the HTTPClient library handling authentication and request formatting efficiently. SMS messages were delivered promptly, and users received clear and readable toll summaries in real-time.

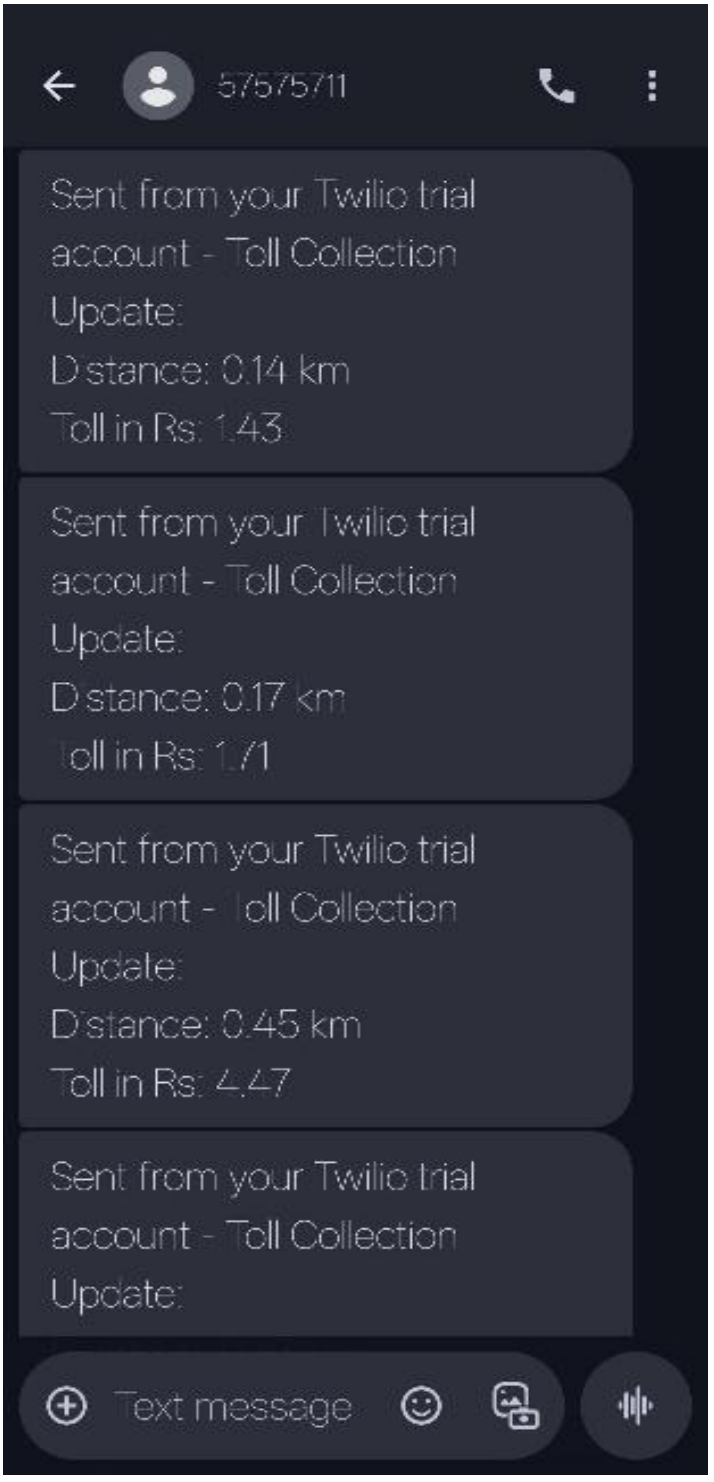
From a usability standpoint, the device provided real-time feedback through the serial monitor, which displayed key information such as current coordinates, segment distances, cumulative distance, and toll values. These outputs not only confirmed the system's internal operationsbut also assistedindebuggingduringthe testingphase. Visual indicators (LEDson the GPSandGSMmodules) provedhelpfulforverifyingmodulereadiness,enhancingoverallеaseof use. In scenarios where the GPS signal was weak or the Wi-Fi connection was interrupted, appropriateerrorhandling messageswеredisplayed, indicating areas forimprovementinterms of network independence and signal reliability.

The testing phase also demonstrated the practical benefits of a distance-based toll system over conventional fixed-point toll booths. Unlike traditional systems that charge uniformly at checkpoints regardless of actual usage, this model ensures users are billed fairly based on how much distance they truly travel. This makes the system particularly advantageous for dynamic or flexible routes and areas with minimal infrastructure.

In summary, the results validate the feasibility of using GPS and internet-based communicationtobuildasmarttollcollectionsystem.Thediscussionsrevealthatwhilethesystem performsreliablyincontrolledconditions,futureiterationscouldbenefit fromenhancementssuch asGSM-basedmessagingtoremoveWi-Fidependency,poweroptimizationforextendedfielduse, and data logging for historical toll records.






5757571
Sent from your Twilio trial account - Toll Collection
Update:
Distance: 0.14 km
Toll in Rs: 1.43
Sent from your Twilio trial account - Toll Collection
Update:
Distance: 0.17 km
Toll in Rs: 1.71
Sent from your Twilio trial account - Toll Collection
Update:
Distance: 0.45 km
Toll in Rs: 4.47
Sent from your Twilio trial account - Toll Collection
Update:
Text message

CHAPTER9

CONCLUSION

In conclusion, the GPS-based automated toll collection system successfully demonstrates the feasibility and effectiveness of using low-cost, portable technologies to create a flexible and scalable solution for toll management. By integrating the ESP32 microcontroller with the NEO-6MGPSmoduleandtheSIM800LGSMmodule, thesystemefficientlytracksvehiclemovement, calculatesdistancetraveled,andsendsreal-timetollupdatesviaSMStousers.Thiseliminatesthe need for fixed toll booths, providing a dynamic, user-friendly approach that can be adapted to varioustransportscenarios,includingprivatevehicles,commercialfleets,andpublictransportation systems. The implementation of the Haversine formula ensures accurate distance calculations, while the threshold mechanism filters out minor GPS noise, improving the system's precision.

The use of the Twilio API to send SMS messages adds an extra layer of convenience, keeping users informed about their toll charges based on actual travel distance. The system's reliability was confirmed through extensive testing, where it demonstrated the ability to provide real-time toll data and respond accurately to user movement. Additionally, the use of Wi-Fi for internetcommunicationenableseasyscalability,allowingthesystemtointegratewithothercloud- based services in the future, such as dynamic toll rate adjustments or location-based billing.

Whilethecurrentimplementationsuccessfullymeetstheproject'sgoals,thesystemcanbe further enhanced by incorporating offline SMS capabilities through GSM, reducing dependency on Wi-Fi. Additionally, further optimization of power usage and the inclusion of a more sophisticated user interface could improve the system's usability for widespread deployment. Overall,thisprojectrepresentsasignificantsteptowardsbuildingamoreintelligent,infrastructure-independent tolling system, leveraging modern GPS and communication technologies to provide an innovative solution for road pricing in both urban and rural environments.

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APPENDIX-A

(PSUEDOCODE)

```
#include <WiFi.h>
#include<HTTPClient.h>
#include <ArduinoJson.h>
#include <TinyGPS++.h>
#include<HardwareSerial.h>

//WiFicredentials
constchar*ssid="wifiplane";
constchar*password="wifiplane1234";

//Twiliocredentials
constchar*twilio_sid="AC9dcab7d7df9fa45584d21aca13d50859";
const char* twilio_token = "d31fa9d028794530d74afbfd41bb1e52";
const char* twilio_number = "+18599035836";
constchar*recipient_number="+917908744849";//Replacewithactualrecipient
number

//Definetollrate
#defineTOLL_RATE50.0

//GPSmodulepins(UART2) #define
GPS_RX_PIN 16
#defineGPS_TX_PIN17

// Minimum movement threshold (in meters)
#defineMIN_MOVEMENT_THRESHOLD0.5
```

```
// Initialize GPS module
HardwareSerialgpsSerial(2);
TinyGPSPlus gps;

// Variables for location tracking
doublelat_prev=0.0,lon_prev=0.0;
double totalDistance = 0.0;
doublelastSentDistance=0.0;//KeepstrackoflastSMS-sentdistance

void setup()
   { Serial.begin(11520
   0);
   gpsSerial.begin(9600, SERIAL_8N1,GPS_RX_PIN,GPS_TX_PIN);

   WiFi.begin(ssid, password);
   Serial.print("ConnectingtoWiFi...");
   while(WiFi.status()!=WL_CONNECTED){ Se
      rial.print(".");
      delay(1000);
   }
   Serial.println("\nConnectedtoWiFi.");
}

voidloop(){
   while(gpsSerial.available()>0){ char
      c = gpsSerial.read(); gps.encode(c);
   }
```

```
if(gps.location.isUpdated()){
    double lat_current = gps.location.lat();
    doublelon_current=gps.location.lng();

    Serial.printf("CurrentLocation:%.6f,%.6f\n",lat_current,lon_current);

    if(lat_prev==0.0&&lon_prev==0.0){ lat_p
        rev = lat_current;
        lon_prev = lon_current;
        Serial.println("Startingpointrecorded.");
    }else {
        double distance = calculateDistance(lat_prev, lon_prev, lat_current,
lon_current);

        //Ignoreupdatesifthesegmentdistanceisbelowthreshold if
        (distance >= MIN_MOVEMENT_THRESHOLD) {
            totalDistance+=distance;
            Serial.printf("Segment Distance: %.2f meters\n", distance);
            Serial.printf("TotalDistance:%.2fmeters\n",totalDistance);

            lat_prev = lat_current;
            lon_prev=lon_current;

            doubletoll=(totalDistance/1000.0)*TOLL_RATE; Serial.printf("Current
            Toll: %.2f\n", toll);

            //SendSMSupdateevery5km
            if(totalDistance-lastSentDistance>=10000){
                StringsmsMessage="TollCollectionUpdate:\nDistance:" +
```

```
                    String(totalDistance/1000.0,2)+ "
                    km\nToll: " + String(toll);
            sendSMS(recipient_number,smsMessage);

            lastSentDistance=totalDistance;
          }
        }else{
          Serial.println("Smallmovementdetected(<0.5m),ignoring...");
        }
      }
    }
}

//Haversineformulatocalculatedistance
doublecalculateDistance(doublelat1,doublelon1,doublelat2,doublelon2){ const
  double R = 6371000;// Earth's radius in meters
  double dLat = radians(lat2 - lat1);
  double dLon = radians(lon2 - lon1);
  doublea=sin(dLat/2)*sin(dLat/2)+
        cos(radians(lat1))*cos(radians(lat2))*sin(dLon/2)*sin(dLon/2); double c
  = 2 * atan2(sqrt(a), sqrt(1 - a));
  returnR* c;
}

  //FunctiontosendSMSusingTwilioAPI
voidsendSMS(Stringnumber,Stringmessage){ if
    (WiFi.status() != WL_CONNECTED)
    { Serial.println("WiFi not connected!");
    return;
```

```
    }

    HTTPClient http;
    Stringurl="https://api.twilio.com/2010-04-01/Accounts/"+String(twilio_sid)+
"/Messages.json";

    StringpostData="To="+number+"&From="+String(twilio_number)+ "&Body="
+ message;

    http.begin(url);
    http.setAuthorization(twilio_sid,twilio_token);
    http.addHeader("Content-Type","application/x-www-form-urlencoded");

    int httpResponseCode = http.POST(postData);

    if (httpResponseCode > 0)
        { Serial.println("SMSSentSuccessfully.")
        ;
    }else{
        Serial.printf("ErrorsendingSMS:%d\n",httpResponseCode);
    }

    http.end();
```

APPENDIX-B

(SCREENSHOTS)

```
#include
<WiFi.h>#include
<HTTPClient.h>#include<
ArduinoJson.h>#include
<TinyGPS++.h>
#include<HardwareSerial.h>

//WiFicredentials
constchar*ssid="wifiplane";
constchar*password="wifiplane1234";

//Twiliocredentials
constchar*twilio_sid="AC9dcab7d7df9fa45584d21aca13d50859";
constchar*twilio_token="d31fa9d028794530d74afbfd41bb1e52";
const char* twilio_number = "+18599035836";
const char* recipient_number ="+917908744849";// Replace withactual recipient
number

// Define toll rate
#defineTOLL_RATE50.0

//GPSmodulepins(UART2)
#define GPS_RX_PIN 16
#defineGPS_TX_PIN17

//Minimummovementthreshold(inmeters) #define
MIN_MOVEMENT_THRESHOLD 0.5

// Initialize GPS module
HardwareSerialgpsSerial(2);
TinyGPSPlus gps;

// Variables for location tracking
doublelat_prev=0.0,lon_prev=0.0;
double totalDistance = 0.0;
doublelastSentDistance=0.0;//KeepstrackoflastSMS-sentdistance

void setup()
    { Serial.begin(115200)
    ;
    gpsSerial.begin(9600,SERIAL_8N1,GPS_RX_PIN,GPS_TX_PIN);
```

```
    WiFi.begin(ssid, password);
    Serial.print("ConnectingtoWiFi...");
```

```
    while(WiFi.status()!=WL_CONNECTED){ Serial.
        print(".");
        delay(1000);
    }
    Serial.println("\nConnectedtoWiFi.");
}

voidloop(){
    while(gpsSerial.available()>0){ char
        c = gpsSerial.read();
        gps.encode(c);
    }

    if(gps.location.isUpdated()){
        doublelat_current=gps.location.lat();
        doublelon_current=gps.location.lng();

        Serial.printf("CurrentLocation:%.6f,%.6f\n",lat_current,
lon_current);

        if(lat_prev==0.0&&lon_prev==0.0){ lat_pre
            v = lat_current;
            lon_prev = lon_current;
            Serial.println("Startingpointrecorded.");
        }else{
            doubledistance=calculateDistance(lat_prev,lon_prev,lat_current,
lon_current);

            //Ignoreupdatesifthesegmentdistanceisbelowthreshold if
            (distance >= MIN_MOVEMENT_THRESHOLD) {
                totalDistance+=distance;
                Serial.printf("Segment Distance: %.2f meters\n", distance);
                Serial.printf("TotalDistance:%.2fmeters\n",totalDistance);

                lat_prev=lat_current;
                lon_prev=lon_current;

                doubletoll=(totalDistance/1000.0)*TOLL_RATE;
                Serial.printf("Current Toll: %.2f\n", toll);

                //SendSMSupdateevery5km
                if(totalDistance-lastSentDistance>=10000){
                    StringsmsMessage="TollCollectionUpdate:\nDistance:"+
                                        String(totalDistance/1000.0,2)+ "
                                        km\nToll: "+ String(toll);
                    sendSMS(recipient_number,smsMessage);

                    lastSentDistance=totalDistance;
                }
```

```
            }else{
                Serial.println("Smallmovementdetected(<0.5m),ignoring...");
            }
        }
    }
}

//Haversineformulatocalculatedistance
doublecalculateDistance(doublelat1,doublelon1,doublelat2,doublelon2){ const
    double R = 6371000;// Earth's radius in meters
    doubledLat=radians(lat2-lat1);
    doubledLon=radians(lon2-lon1);
    doublea=sin(dLat/2)*sin(dLat/2)+
              cos(radians(lat1))*cos(radians(lat2))*sin(dLon/2)*sin(dLon
/2);
    doublec=2*atan2(sqrt(a),sqrt(1-a)); return
    R * c;
}

//FunctiontosendSMSusingTwilioAPI
voidsendSMS(Stringnumber,Stringmessage){ if
    (WiFi.status() != WL_CONNECTED) {
        Serial.println("WiFinotconnected!");
        return;
    }

    HTTPClienthttp;
    Stringurl="https://api.twilio.com/2010-04-01/Accounts/"+ String(twilio_sid)
+ "/Messages.json";

    StringpostData="To="+number+"&From="+String(twilio_number)+ "&Body="+
message;

    http.begin(url);
    http.setAuthorization(twilio_sid,twilio_token);
    http.addHeader("Content-Type", "application/x-www-form-urlencoded");

    int httpResponseCode = http.POST(postData);

    if (httpResponseCode >0)
        { Serial.println("SMSSentSuccessfully.");
    }else{
        Serial.printf("ErrorsendingSMS:%d\n",httpResponseCode);
    }

    http.end();
}
```

APPENDIX-C

ENCLOSURES

PaperPublicationsRelatedToTheProjectworkinJournals

Certificates






DOI: 10.55041/IJSREM48786
IJSREM
e-Journal
ISSN: 2582-3930
Impact Factor: 8.586
INTERNATIONAL JOURNAL OF SCIENTIFIC RESEARCH IN ENGINEERING & MANAGEMENT
An Open Access Scholarly Journal || Index in major Databases & Metadata
CERTIFICATE OF PUBLICATION
International Journal of Scientific Research in Engineering & Management is hereby awarding this certificate to
Bhavya
in recognition to the publication of paper titled
GPS Based Toll Collection System Using ESP32
published in IJSREM Journal on Volume 09 Issue 05 May, 2025
Editor-in-Chief
IJSREM Journal
www.ijsrem.com
e-mail: editor@ijsrem.com


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e-Journal
ISSN: 2582-3930
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Editor-in-Chief
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